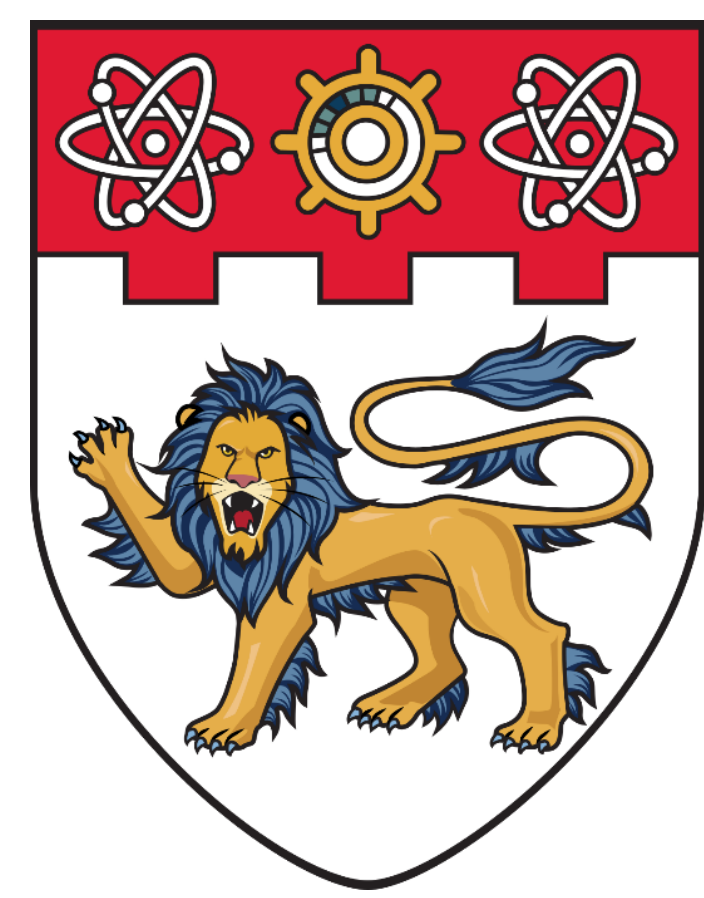
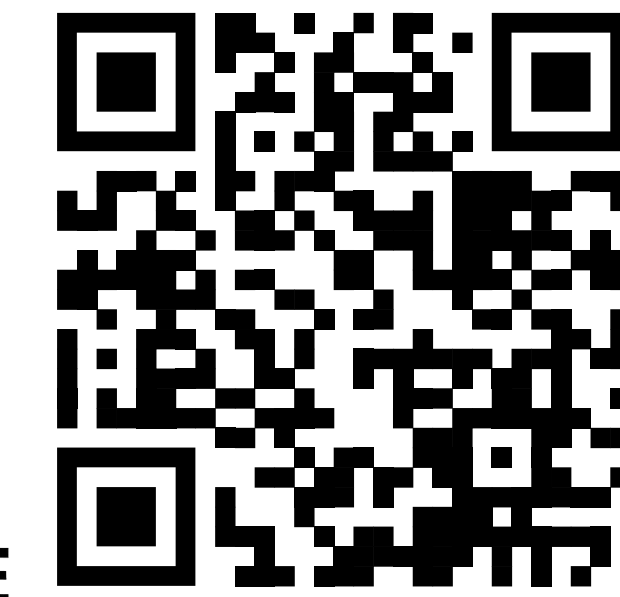


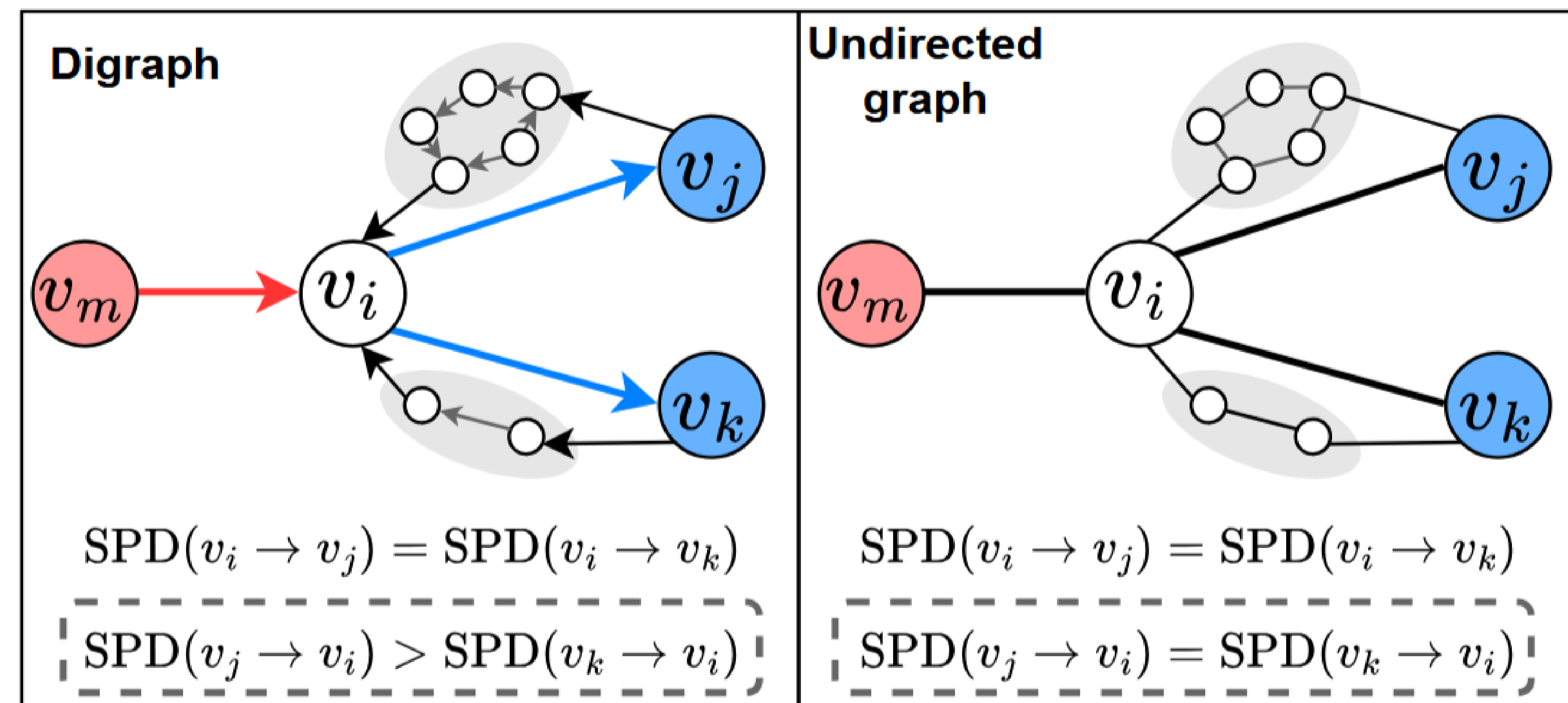


Commute Graph Neural Networks

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Background and Motivation



Directed Graphs
 Unidirectional SPDs are equal does not imply that mutual SPDs will also be equal.

Undirected Graphs
 Shortest path distances (SPD) between nodes are symmetric.

- Q1: How crucial is it to retain random walk distances in graph learning?

Hitting Time reflects the directed influence or connectivity.
Commute Time offers insights into the mutual relationships between nodes, such as celebrity-follower relationships.

- Q2: Are message-passing GNNs capable of preserving random walk distances?

Hitting Time
Commute Time

GOAL

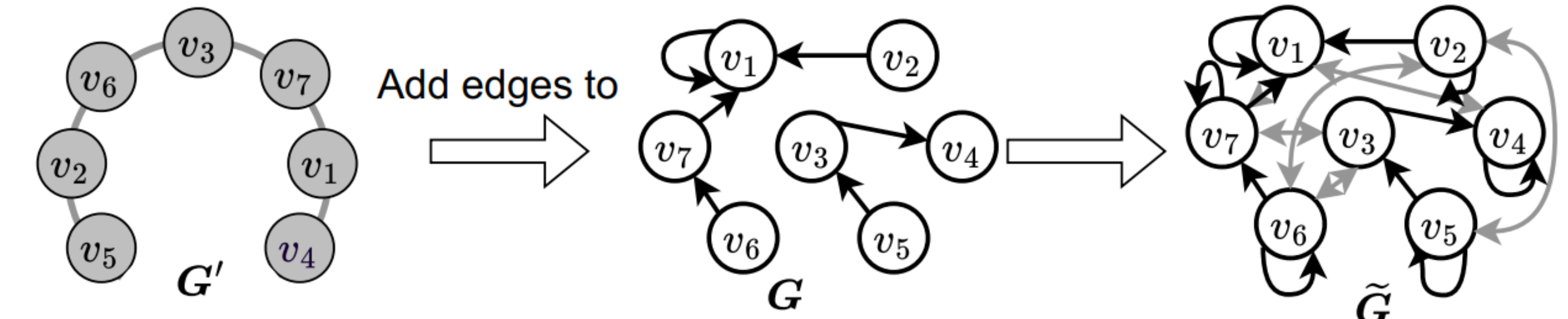
Directly encode node-wise commute time into the node representations to accurately reflect the true interaction strength between *adjacent* nodes during neighbor aggregation, accounting for both forward and backward paths.

Method

- Digraph Laplacian (DiLap):** the divergence of the gradient on digraphs.

$$\mathbf{T}s = \mathcal{G}\mathcal{D}s, \quad \mathbf{T} = \mathbf{B} \text{diag} \left(\{\mathbf{P}_{ij}\}_{(v_i, v_j) \in E}^M \right) \mathbf{B}^\top$$
- Similarity-based Graph Rewiring:** make a given graph irreducible and aperiodic.

$$\Omega = \{v_5, v_2, v_6, v_3, v_7, v_1, v_4\}$$



- From DiLap to Commute Time Graph Neural Networks (CGNN):**

Theorem: The hitting time and commute time from v_i to v_j can be computed as:

$$h(v_i, v_j) = \frac{\tilde{\mathbf{T}}_{jj}^\dagger}{\pi_j} - \frac{\tilde{\mathbf{T}}_{ij}^\dagger}{\sqrt{\pi_i \pi_j}},$$

$$c(v_i, v_j) = \frac{\tilde{\mathbf{T}}_{jj}^\dagger}{\pi_j} + \frac{\tilde{\mathbf{T}}_{ii}^\dagger}{\pi_i} - \frac{\tilde{\mathbf{T}}_{ij}^\dagger}{\sqrt{\pi_i \pi_j}} - \frac{\tilde{\mathbf{T}}_{ji}^\dagger}{\sqrt{\pi_i \pi_j}}.$$

CGNN

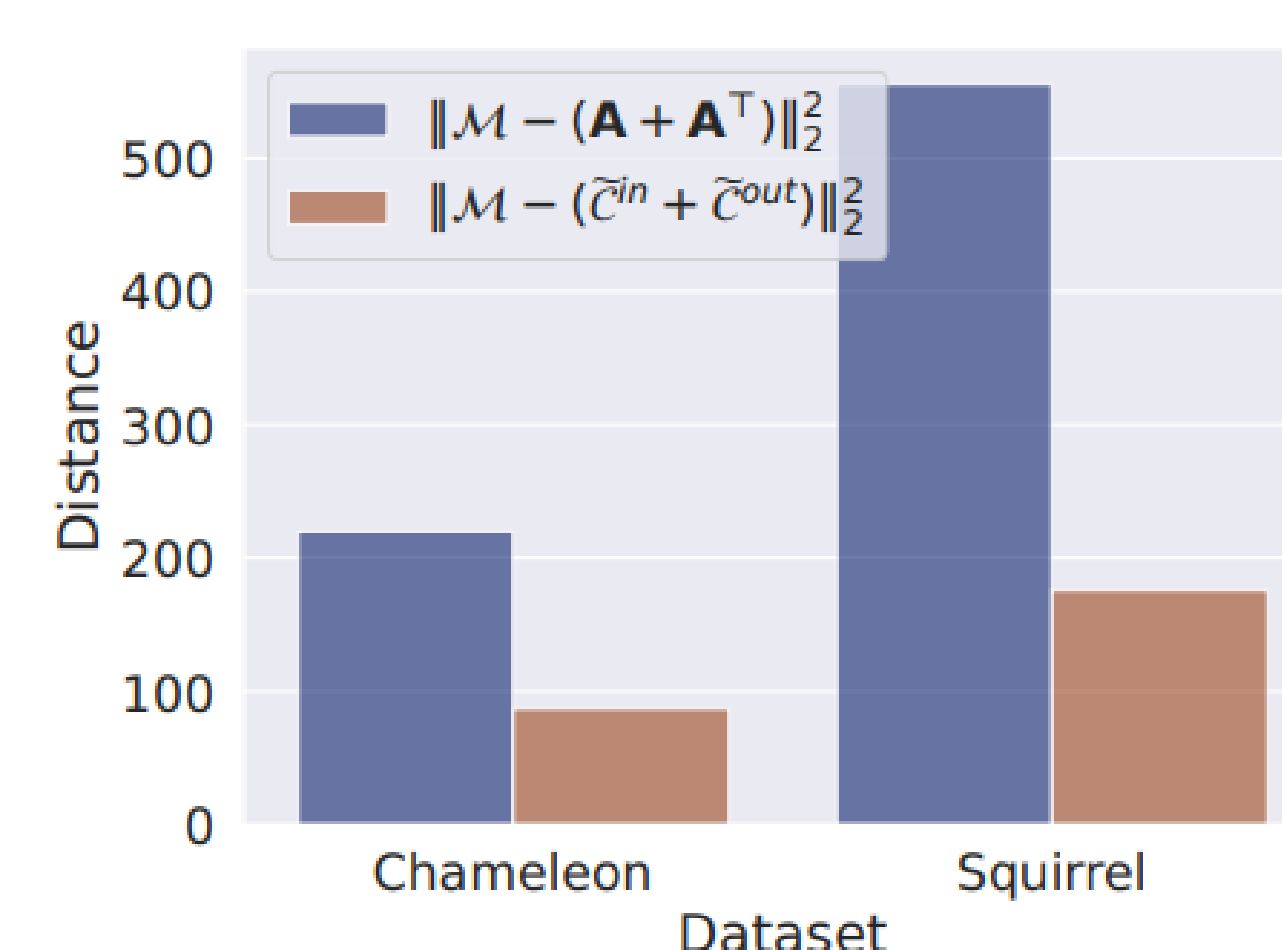
$$m_{i, \text{in}}^{(\ell)} = \text{Agg}_{\text{in}}^{(\ell)} \left(\left\{ \tilde{\mathcal{C}}_{ij}^{\text{in}} \cdot h_j^{(\ell-1)} : v_j \in \mathcal{N}_i^{\text{in}} \right\} \right)$$

$$m_{i, \text{out}}^{(\ell)} = \text{Agg}_{\text{out}}^{(\ell)} \left(\left\{ \tilde{\mathcal{C}}_{ij}^{\text{out}} \cdot h_j^{(\ell-1)} : v_j \in \mathcal{N}_i^{\text{out}} \right\} \right)$$

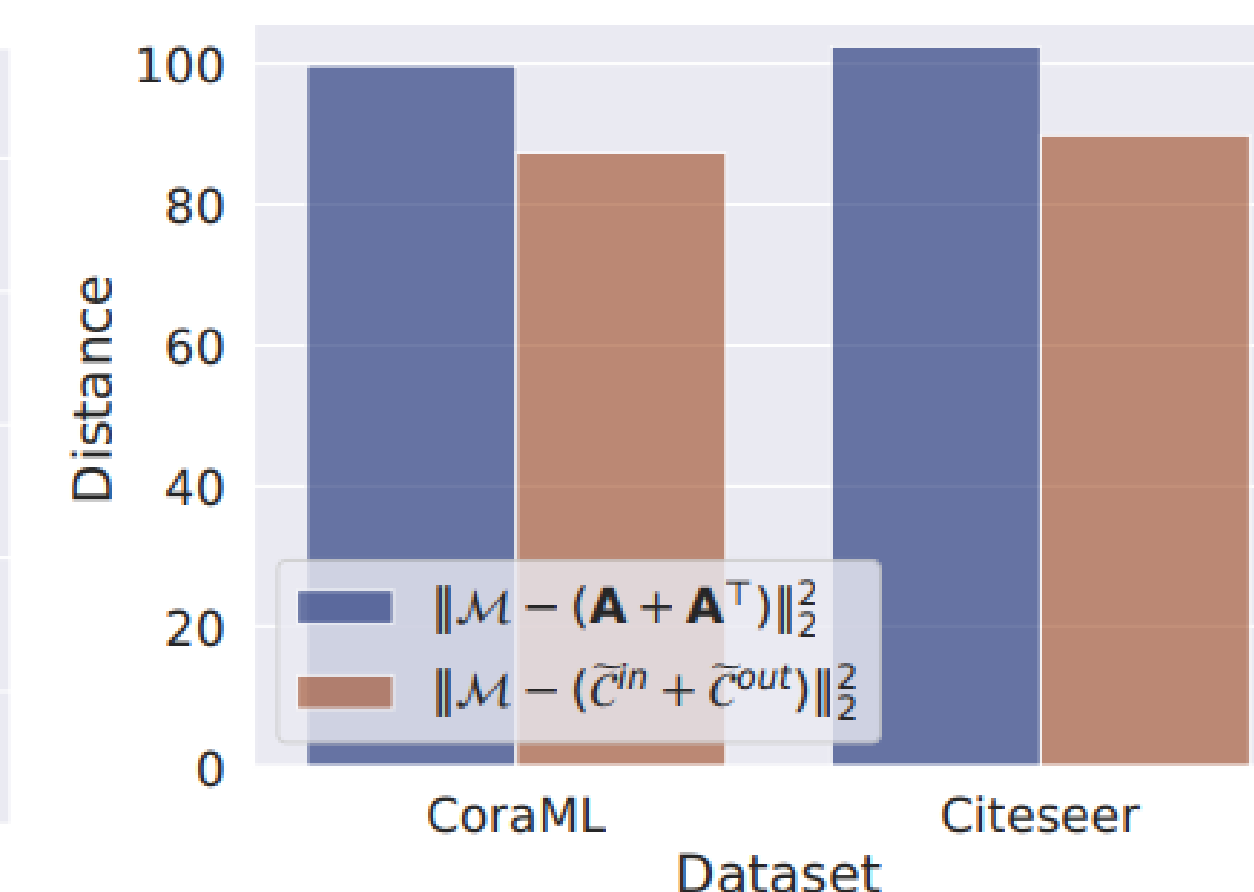
$$h_i^{(l)} = \text{Comb}^{(\ell)} \left(h_i^{(\ell-1)}, m_{i, \text{in}}^{(\ell)}, m_{i, \text{out}}^{(\ell)} \right)$$

Experiments

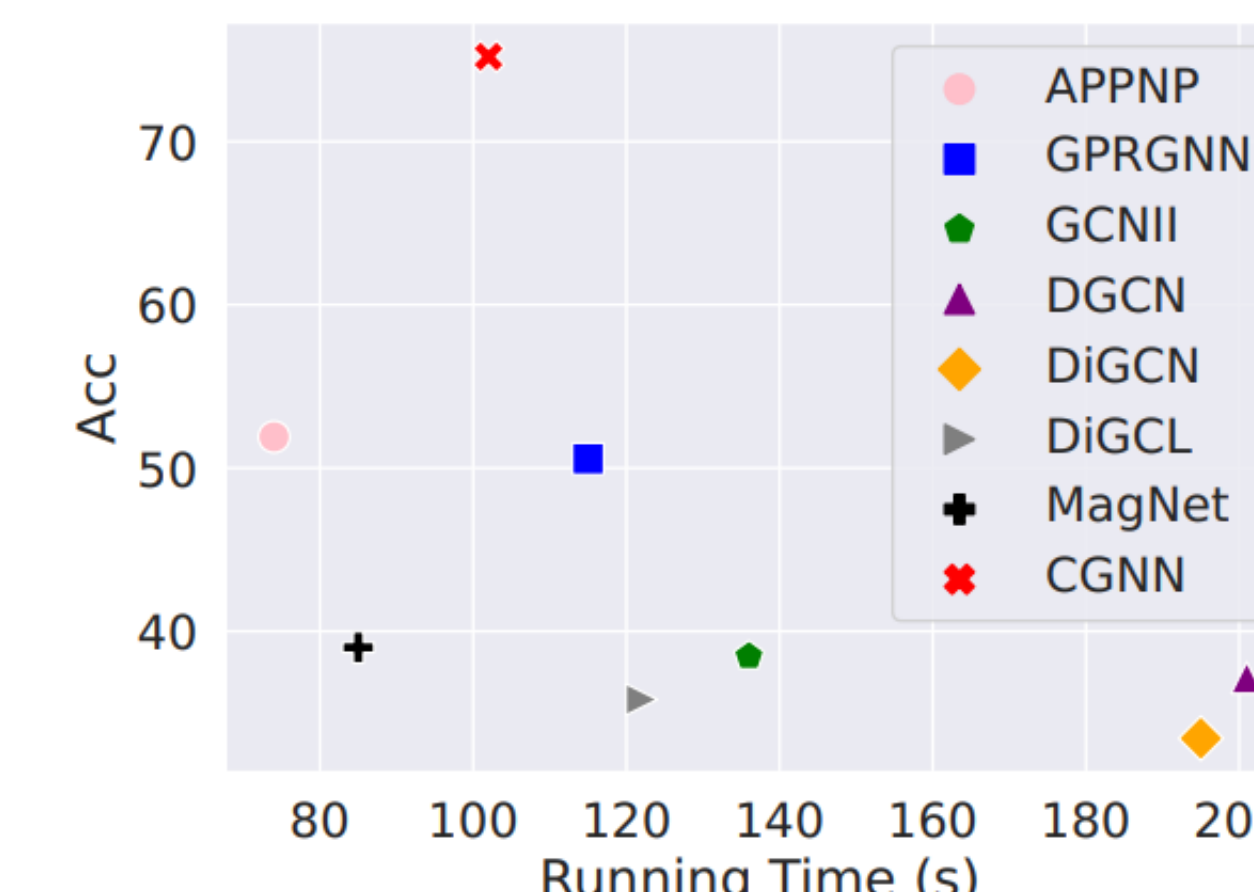
Method	Squirrel	Chameleon	Citeseer	CoraML	AM-Photo	Snap-Patents	Roman-Empire	Arxiv-Year
GCN	52.43±2.01	67.96±1.82	66.03±1.88	70.92±0.39	88.52±0.47	51.02±0.06	73.69±0.74	46.02±0.26
GAT	40.72±1.55	60.69±1.95	65.58±1.39	72.22±0.57	88.36±1.25	OOM	49.18±1.35	45.30±0.23
GraphSAGE	41.61±0.74	62.01±1.06	66.81±1.38	74.16±1.55	89.71±0.57	67.45±0.53	86.37±0.80	55.43±0.75
APPNP	51.91±0.56	45.37±1.62	66.90±1.82	70.31±0.67	87.43±0.98	51.23±0.54	72.96±0.38	50.31±0.42
MixHop	43.80±1.48	60.50±2.53	56.09±2.08	65.89±1.50	87.17±1.34	41.22±0.19	50.76±0.14	45.30±0.26
GPRGNN	50.56±1.51	66.31±2.05	61.74±1.87	73.31±1.37	90.23±0.34	40.19±0.03	64.85±0.27	45.07±0.21
GCNII	38.47±1.58	63.86±3.04	58.32±1.93	64.84±0.71	83.40±0.79	48.09±0.09	74.27±0.13	57.36±0.17
DGCN	37.16±1.72	50.77±3.31	66.37±1.93	75.02±0.50	87.74±1.02	OOM	51.92±0.43	OOM
DiGCN	33.44±2.07	50.37±4.31	64.99±1.72	77.03±0.70	88.66±0.51	OOM	52.71±0.32	48.37±0.19
MagNet	39.01±1.93	58.22±2.87	65.04±0.47	76.32±0.10	86.80±0.65	OOM	88.07±0.27	60.29±0.27
DUPLEX	57.60±0.98	61.25±0.94	67.60±0.72	72.26±0.71	87.80±0.82	66.54±0.11	79.02±0.08	64.37±0.27
DiGCL	35.82±1.73	56.45±2.77	67.42±0.14	77.53±0.14	89.41±0.11	70.65±0.07	87.94±0.10	63.10±0.06
DirGNN	75.19±1.26	79.11±2.28	66.57±0.74	75.33±0.32	88.09±0.46	73.95±0.05	91.23±0.32	64.08±0.26
CGNN	77.83±1.52	79.62±2.33	71.59±0.16	77.08±0.54	90.42±0.10	72.89±0.24	92.87±0.45	66.16±0.32



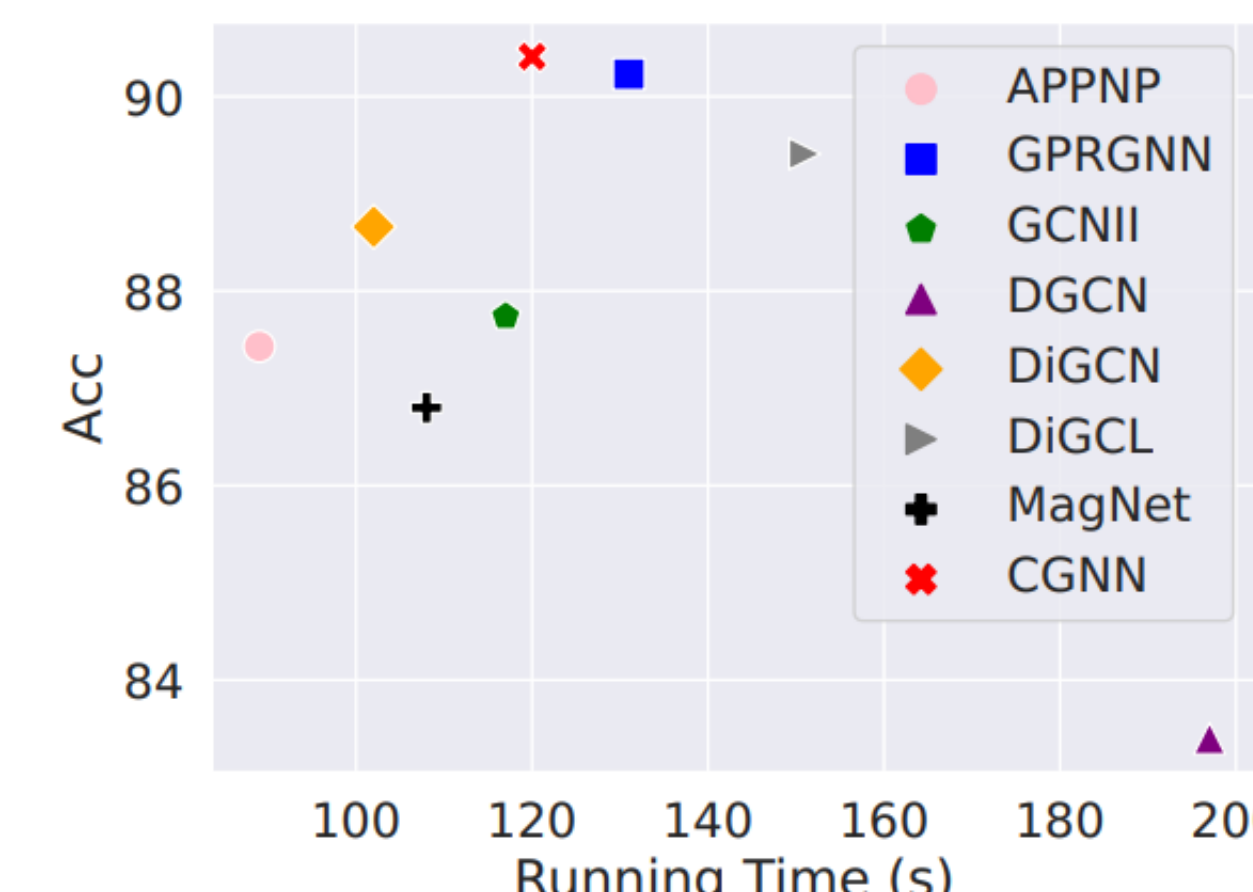
(a) Heterophilic graph.



(b) Homophilic graph.



(a) Squirrel



(b) AM-Photo

- CGNN achieves new state-of-the-art results on 6 out of 8 datasets.
- Commute times effectively filters out irrelevant information by appropriately weighting neighbors.
- CGNN gets the best trade-off between effectiveness and efficiency.